

AI CRYPTO SIGNAL AGENT

CHECKPOINT 09R1

REAL TELEGRAM DELIVERY

FINAL CLOSURE

CP-09R1-CLOSED

Exactly one owner-authorized production signal delivered to Telegram,
then forensically proven without retry or production mutation.

AUTHORITATIVE REMEDIATION RELEASE

d7218fbc443057e64ae28c603c55fc599dd7909e

Status: PHASE 09R1 PASS - EVIDENCE SEALED - NO RETRY

Checkpoint Identity

Current Phase	Phase 09R1 - Production Signal Telegram Delivery Validation
Current Checkpoint	CP-09R1-CLOSED
Final Milestone	Exactly one real production signal delivered to Telegram and forensically proven
Final Phase Status	PASS
Delivery Mode	Exactly one owner-authorized static one-shot
Remediation Branch	remediation-phase09r1-from-phase08
Remediation HEAD	d7218fbc443057e64ae28c603c55fc599dd7909e
Forensic Master HEAD	cfac13d35868a4b18e3670b56822c972675c7417
Installed Release	/opt/ai-crypto-signal-agent-releases/d7218fbc443057e64ae28c603c55fc599dd7909e
Installed Unit SHA-256	a3def93c11fb727427e9c467584b0d1093c7c113dc88a55285534c51d4fa1d6f
Checkpoint Classification	FINAL PHASE 09R1 GOVERNANCE CLOSURE - EVIDENCE SEALED
Remote Lock	Pending separate repository integration and push authorization

Final Verdict

PASS - PHASE 09R1 REAL TELEGRAM DELIVERY FORENSICALLY PROVEN WITHOUT RETRY

Phase 09R1 is formally closed as a successful real Telegram-delivery phase. The sole authorized live execution initially ended in a conservative HOLD because the immediate report lacked a new evaluation artifact and a retained systemd InvocationID. A later read-only forensic audit proved that the published-signal contract correctly persists a publication rather than a separate evaluation, uniquely recovered the InvocationID, and bound the validated Telegram receipt to the sole authorized invocation. No retry or production mutation was used.

Locked Operating Boundary

Boundary	Locked state
Execution authority	Exactly one owner-authorized live runner invocation and one systemctl start --no-block.
Automatic activation	Disabled: no timer, cron, path, socket, daemon loop, or enablement path.
Retry policy	No automatic retry. No second live attempt authorized or required.
Trading execution	Disabled. This checkpoint proves signal publication and Telegram delivery only.
Service mode	Static systemd Type=oneshot with Restart=no.
Downstream work	Repository integration, remote lock, TRUSTED_CP09 binding, and Phase 10-12 reconciliation remain separate.

Phase Objective and Closure Timeline

Objective

Demonstrate exactly one real Phase 09R1 production signal delivered to Telegram under a manual, fail-closed, one-shot operating contract with quota acquisition, reservation release, sanitized evidence, and no trading execution.

Authoritative Timeline

Date	Milestone	Result
2026-07-26	Initial live start	Failed before Python with systemd 203/EXEC; no network or runtime mutation.
2026-07-26	Replacement one-shot	Exited 7 before publication, quota, or Telegram; exposed insufficient exit-7 observability.
2026-07-26	Observability validation	Exited 7 with sanitized ELIGIBLE_SETUP_CONSTRUCTION TypeError evidence.
2026-07-26	Offline forensic audit	Root cause identified: pandas Timestamp was not normalized before deterministic hashing.
2026-07-26	Timestamp remediation	Commit d7218fbc normalized supported timestamp values; full regression passed.
2026-07-27	Runner V5 and readiness	External readiness-seal contract, one-execution live authorization, and pre-live readiness passed.
2026-07-27	Sole live one-shot	Service returned success with valid Telegram receipt; immediate verdict remained conservative HOLD.
2026-07-27	Read-only forensic closure	Exactly one attempt and current receipt binding proven; InvocationID recovered uniquely.
2026-07-27	Final governance closure	Phase 09R1 formally closed as PASS; another live attempt prohibited.

Closure Decision

FORMALLY_CLOSE_PHASE09R1_AS_SUCCESSFUL_REAL_TELEGRAM_DELIVERY_PHASE

Release, Unit, Runner and Readiness Identities

Remediation commit	d7218fbc443057e64ae28c603c55fc599dd7909e
Installed release	/opt/ai-crypto-signal-agent-releases/d7218fbc443057e64ae28c603c55fc599dd7909e
Installed unit	/etc/systemd/system/ai-crypto-signal-agent.service
Installed unit SHA-256	a3def93c11fb727427e9c467584b0d1093c7c113dc88a55285534c51d4fa1d6f
V5 runner SHA-256	2fbc0e0b335d35cd7d03fa1553439b4214b334742d96311b1e2fcb2c53f846e3
V5 logic SHA-256	5460ad8c08d2fcf9e1c84d26be02bd3ffaf3225a4bc0d62e8d0594e1f1d172b5
V5 contract SHA-256	84322db623c90235bdaafc801c930dcc1180c4cee1308249bf4d2b137421af9a
V5 identity SHA-256	02b901a51b1c5eca9c6f5803e29cb37907ce512c6e0b185a882f36684691b4b1
V5 evidence manifest	a62edaaaac933cd4bfe86f28f5a811c89d14c519bca9e4c2ba26e72bf494f83a
Readiness manifest	76061a1964a1359b1f0694421020026adda71d6c81de6b9bcc121536530b42ac
Readiness seal SHA-256	b5d76037dcd90dec2951085ac874783de1f90fa4e5f029caccee3352de88eb4f

Static Unit Contract

Control	Verified value
UnitFileState	static
Type	oneshot
Restart	no
ExecStart count	1
Automatic activation paths	0
Daemon reload during closure	Not executed
Installed release mutation	No
Credential values embedded	No

Runner Safety Contract

- Exactly one future systemctl start --no-block path; no second-start authority.
- No restart, try-restart, stop, reset-failed, daemon-reload, retry loop, or automatic rollback path.
- External readiness seal grants no live authority; separate ten-key live authorization was mandatory.
- Runner evidence root remains immutable and is not mutated by pre-live readiness or live execution.
- Classifier PYTHONPATH is explicitly bound to the installed immutable release.

Sole Live Execution Evidence

Owner authorization	AUTHORIZE_EXACTLY_ONE_PHASE09R1_TIMESTAMP_NORMALIZATION_VALIDATION_LIVE_ONES HOT
Live authorization SHA-256	1ffaca747cd79665f8990603b80ef9bd467bc6f6da02a2e4595f8ca8bd13063c
Future guard SHA-256	a8d4f7bef1378cf1f7673f5318ab3740c48f49ee0e8640fdef9f12750edf92d3
Runner invocation count	1
Second runner invocation count	0
systemctl start count	1
NRestarts	0
Runner exit code	0
Unit result / ExecMainStatus	success / 0
Recovered systemd InvocationID	b799f636028548b0ac5cce0c81867763

Production Outcome

Evidence item	Verified result
Sanitized outcome	DELIVERY_SUCCEEDED_WITH_VALID_RECEIPT
Telegram boundary reached	YES
Telegram receipt valid	YES
Telegram message_id present	YES
Telegram attempt minimum / maximum	1 / 1
Exactly one Telegram attempt proven	YES
Current invocation receipt binding	PASS
New publication count	1
New production outcome count	1
Quota acquire / increment count	1 / 1
Reservation final state	RELEASED
Residual publication lock count	0

TARGET_ONE_REAL_TELEGRAM_SIGNAL_PROVEN = YES

Verdict Reconciliation and Forensic Findings

Immediate Live Verdict

HOLD_TELEGRAM_DELIVERY_AMBIGUOUS_NO_RETRY

The immediate report correctly preserved a conservative HOLD because no new evaluation artifact was detected and the current systemd InvocationID was unavailable after exit. Because the Telegram boundary had been reached, the locked policy prohibited retry.

Superseding Read-Only Forensic Verdict

PASS_PHASE09R1_REAL_TELEGRAM_DELIVERY_FORENSICALLY_PROVEN_WITH_OUT_RETRY

Finding	Forensic conclusion	Classification	Closure effect
Evaluation artifact	Not expected for a published signal. Production contract persists the publication and reserves evaluation artifacts for NO_TRADE.	EVALUATION_CONTRACT_MISMATCH	Non-blocking
InvocationID collection	V5 captured the active InvocationID, then overwrote it with an empty post-exit value before journal collection.	POST_RUN_COLLECTION_DEFECT	Non-blocking

Recovered Invocation Identity

The InvocationID b799f636028548b0ac5cce0c81867763 was uniquely bound through the exact unit, boot ID, service PID 896346, runner-observed MainPID, and the sole-run timeline. The recovered journal was scopable to that unique InvocationID. No InvocationID was invented from timestamp proximity alone.

Reconciliation Rule

The original live HOLD remains preserved as the authoritative immediate classification. The forensic PASS is a superseding closure determination based on additional read-only evidence; it does not rewrite the live evidence and does not authorize another attempt.

Authority Consumption, No-Retry Proof and Preservation

Authority control	Final state
Historical live guards	3 present and unchanged
Timestamp-normalization live authorization	Created once and consumed
Timestamp-normalization future guard	Created once; never deleted or reused
Second start authority	Does not exist
Automatic retry authorized	NO
Another Phase 09R1 live attempt authorized	NO
Another Phase 09R1 live attempt required	NO

Preservation Audit

Protected object	Verified preservation
Production runtime	Unchanged during forensic and closure audits
Live and forensic evidence	Unchanged; manifests verified
V5 runner and evidence	Unchanged
Readiness seal and evidence	Unchanged
Live authorization and guards	Unchanged after consumption
Installed unit and release	Unchanged
Repositories	Unchanged; no push
Service operations during closure	None
Network operations during closure	None
Credential contents	Not read or printed

Safety Status at Closure

- Trading execution remains disabled.
- Automatic activation remains disabled.
- Automatic retry remains disabled.
- The service remains a controlled static one-shot boundary.
- No rollback, replay, or replacement signal is authorized.

Evidence Chain and Closure Package

Evidence package	Authoritative root	Manifest SHA-256
Runner V5 evidence	/opt/ai-crypto-signal-agent-forensics/phase09r1-runner-external-readiness-seal-v5-20260727T074317Z	a62edaaaac933cd4bfe86f28f5a811c89d14c519bca9e4c2ba26e72bf494f83a
Pre-live readiness V2 retry	/opt/ai-crypto-signal-agent-forensics/phase09r1-pre-live-readiness-v2-retry-20260727T081056Z	872c8a5ff1a57aaf77ac4baa1b4a59ce508274b611d8d2ac57d257e72fa32461
Sole live execution	/opt/ai-crypto-signal-agent-forensics/phase09r1-timestamp-normalization-validation-live-20260727T082555Z	845781d2d3c06ba0588aba953b2ee0189164d972492082121e096efa19cf92d4
Post-live forensic audit	/opt/ai-crypto-signal-agent-forensics/phase09r1-post-live-ambiguous-delivery-forensic-20260727T084851Z	349e5c4f88b4668b7af33670ad2a7e08992bae5d85304eed4b32aa95371f627f
Final closure	/opt/ai-crypto-signal-agent-forensics/phase09r1-final-closure-20260727T090945Z	73d62eea618cb4f48d74f8eaf30335aa49f56680b61c24365cbdba94cde382a3

Final Closure Package

- owner-closure-decision.txt
- phase09r1-final-status.txt
- verdict-reconciliation.txt
- delivery-proof-summary.txt
- evaluation-contract-reconciliation.txt
- observability-findings.txt
- authority-consumption-summary.txt
- repository-release-status.txt
- checkpoint-source-record.txt
- preservation-report.txt
- final-report.txt
- sealed-summary.txt
- evidence-sha256-manifest.txt

Closure Evidence Identity

Evidence root	/opt/ai-crypto-signal-agent-forensics/phase09r1-final-closure-20260727T090945Z
Manifest entries	15
Manifest SHA-256	73d62eea618cb4f48d74f8eaf30335aa49f56680b61c24365cbdba94cde382a3

Final Closure Status and Pending Governance Work

Closure item	Status
Phase 09R1 final status	PASS
One real Telegram signal proven	YES
Exactly one Telegram attempt proven	YES
Delivery ambiguity remaining	NO
Another live attempt authorized	NO
Checkpoint PDF	CREATED BY THIS DOCUMENT
Remote lock completed	NO
TRUSTED_CP09_COMMIT bound	NO
Phase 12 work executed in closure	NO

Pending Separate Authorizations

- Owner review and acceptance of this checkpoint PDF.
- Repository integration and commit-chain scope audit.
- Normal non-force push and remote alignment lock.
- Binding of TRUSTED_CP09_COMMIT to the final integrated repository identity.
- Compatibility and compliance reconciliation of downstream Phase 10-12 work against the trusted Phase 09R1 baseline.

Important Governance Boundary

This checkpoint closes the Phase 09R1 delivery target. It does not itself authorize repository integration, push, TRUSTED_CP09 binding, Phase 12 execution, another live signal, or any automatic retry.

CP-09R1-CLOSED

PHASE 09R1 FORMALLY CLOSED - REAL TELEGRAM DELIVERY PROVEN - NO RETRY

Source basis: owner-provided final closure record referencing checkpoint-source-record.txt and closure manifest 73d62eea618cb4f48d74f8eaf30335aa49f56680b61c24365cbdba94cde382a3.